

AIMO Proof Pilot — Final Grade

Problem (#_ID): 2_ALTORO

Submission: model_2

Scoring scale (0/1/6/7) — two binary decisions:

Step 1. Is the solution essentially complete?

- If **no**: award **1** for significant partial progress, otherwise **0**.
- If **yes**: award **7**, or **6** if there is a minor error or omission.

Grade against the **markscheme** (docs/markschemes.pdf).

1. Final mark

Final mark (choose one):

2. Reasoning

Justify the mark with explicit reference to the markscheme points (e.g. ME1) where possible.

A complete, correct coordinate solution. It sets up admissible coordinates, computes H , the altitude feet, X , and the second circumcircle intersection Q exactly, identifies P as the midpoint of HQ via the nine-point-circle homothety at H , and computes PE exactly to obtain $PQ/PE = 125/91$, hence $a + b = 216$. The only omission — not explicitly ruling out a second intersection of segment HQ with the nine-point circle before naming P — is expressly non-penalised under NME3. Scores **7**.